11. Overview and concepts

Outlier detection

Outliers (or anomalies), by definition, are subsets of data that "do not fit well" with the rest of the data. There are many methods for deciding the manner in which outliers "do not fit well" with the rest of the data. These methods are known as outlier detection algorithms.

There are at least two reasons why you might be interested in detecting outliers in the course of data analysis:

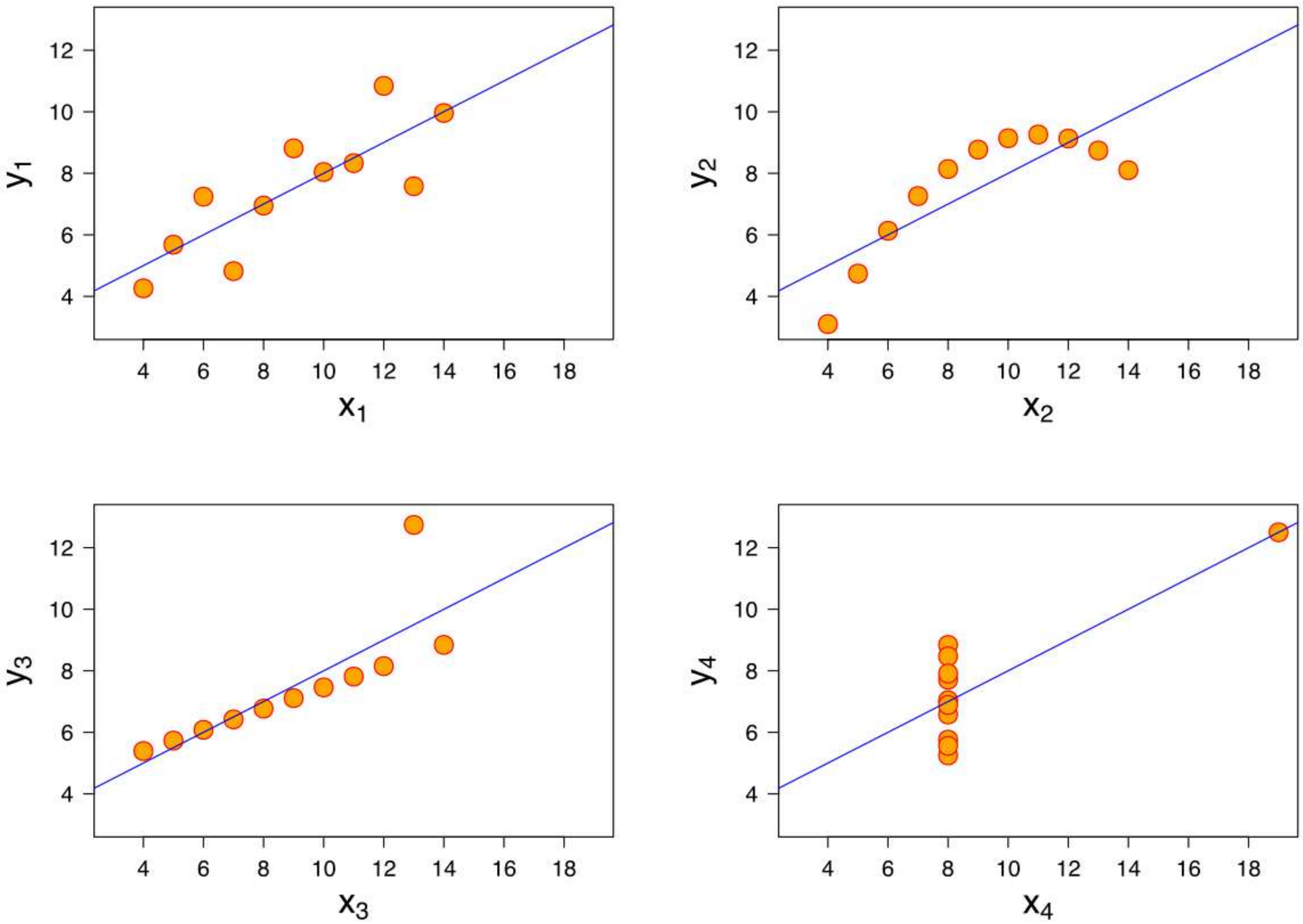
1. Firstly, outliers may represent an extraordinary, unexpected, or unusual phenomenon of certain significance (e.g. credit card fraud, manufacturing defects, etc.).
2. Secondly, outliers in data may adversely affect the quality of a model and therefore must be detected and removed before a model is built. Our lesson in this unit is more concerned with these applications.

Examples from the Anscombe's Quartet

The figure below illustrates Anscombe's Quartet, which is a set of 4 datasets with seemingly identical descriptive statistics:

- Mean of x: 9
- Sample variance of x: 11
- Mean of y: 7.50
- Sample variance of y: 4.125
- Correlation between x and y: 0.816
- Linear regression line: y = 3.00 + 0.500x
- Coefficient of determination of the linear regression: 0.67

When visualised, however, the four datasets possess contrastingly different distributions, as shown in this figure:



The Anscombe's Quartet. Image from Wikipedia [(https://en.wikipedia.org/wiki/Anscombe%27s_quartet)](https://en.wikipedia.org/wiki/Anscombe%27s_quartet).

While Anscombe's Quarter is frequently used to illustrate the importance of visualising your data, it also serves as an excellent example of the negative effect that an outlier in data could exert on a model (e.g. linear regression).

The scatterplot at the bottom left of the above figure indicates the presence of an outlying data point. The presence of the outlier is strong enough to skew an otherwise well-fitted linear regression line (the blue line). Consequently, removing this outlier is expected to improve the correlation score between both variables to approximately 1.0 (perfect positive correlation) and correct the linear regression line.

IQR Method

The simplest method for detecting outliers is the Inter-Quartile Range (IQR) method. The IQR method defines an outlier in a variable as:

$$\text{outlier} < Q_1 - 1.5 \times IQR$$

or

$$\text{outlier} > Q_3 + 1.5 \times IQR$$

, where Q_1 and Q_3 are the 25th percentile and the 75th percentile of all values in the variable, and IQR is the difference between Q_3 and Q_1.

Example

The dataset below contains longevity information (typical lifespan) of 40 different mammal species. Is the elephant an outlier according to its longevity?

MammalLongevity.csv

Source: lock5stat.com (https://www.lock5stat.com/datapage2e.html)

To find the answer, perform these calculations:

1. In the longevity variable, we first determine Q_1=8.00 and Q_3=15.25, thus IQR = Q_3 - Q_1=15 - 8 = 7.25.
2. Multiplying the IQR by 1.5 gives 10.875, which we then use to derive the outlier lower limit as 8 - 10.875 = -2.875 years and the outlier upper limit as 15 + 10.875 = 26.125 years.
3. Since the elephant is the only mammal in the dataset with a longevity of 40 years that is greater than the specified outlier upper limit, it is an outlier.

Solution

MammalLongevity.xlsx